

Monaco 2024 - Lap in Top 10 Classification

Project Goal

The goal of this project is to determine whether lap data and weather conditions can be used to identify whether a given lap was among the Top 10 during the 2024 Formula 1 Monaco Grand Prix. This task involves binary classification: a lap was either in the Top 10 or it was not. In addition to testing the hypothesis, an important part of the work involves comparing three classification models and analyzing which lap characteristics distinguish a driver in the Top 10 from those outside the Top 10.

Data and Approach

The data for the project was sourced from the FastF1 Python library. From there, we utilized lap-specific data, including the driver, speeds on different sections of the track, tyre type, and much more. In addition, we also included data on the weather conditions at the track on that day: air temperature, track temperature, wind speed, and other weather conditions. During data analysis, all laps for which there was no accurate information, laps during which a pit stop occurred, and all variables irrelevant to the problem - such as the race start date - were filtered out. After cleaning, the weather data was combined with the lap data.

As a result of the analysis, three models were selected as the most suitable for the dataset's constraints: Logistic Regression, Random Forest, and Support Vector Machine. Additionally, one of the main criteria was that, given the task at hand, misclassifying a lap in the Top 10 as not in the Top 10 is a more serious error than misclassifying a lap not in the Top 10 as being in the Top 10.

Main Findings

The model results showed that lap data can indeed be used to effectively distinguish the Top 10 laps from the rest. When based solely on performance, the SVM performed best - both in terms of the number of laps identified in the Top 10 and in terms of the balance between correctly identifying laps in the Top 10 and correctly identifying laps outside the Top 10 - although the difference compared with Random Forest was small.

The analysis of feature selection led to the conclusion that not all relatively weak predictors should be removed; sometimes they are beneficial precisely when combined with others. Also, it is not surprising that speed-related variables were among the most useful predictors, but it is interesting that high speed does not always guarantee success; the main predictor - namely, speed at the second speed trap - for the Top 10 is, on average, lower than for drivers outside the Top 10.

Conclusion

Although SVM yields the best results in terms of metrics, it falls short of the others in terms of interpretability; therefore, I consider Random Forest to be the optimal solution, given its balance of quality and explainability. It is also worth noting that there are limitations, as the entire analysis is based on a single race; the determining factors may not necessarily be the same on a different track.